

Instructions to the students

BSCCS2001:DBMS

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1 Introduction

This document contains important information regarding the DBMS course. We will keep updating this document with additional inputs as and when needed. We strongly recommend that you read this document carefully.

2 PostgreSQL

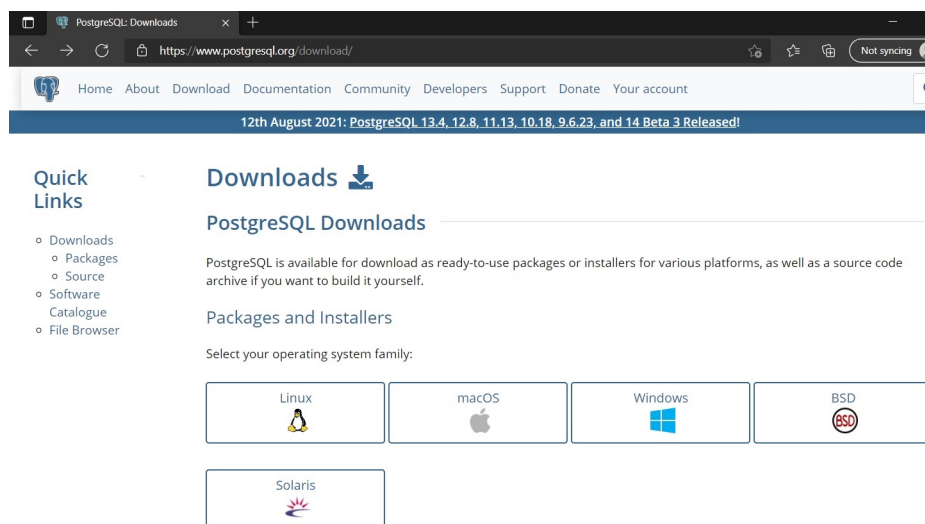
In this course, we use the PostgreSQL database considering many factors including open source licensing (<https://opensource.org/licenses/postgresql>) and the popularity of the database (<https://db-engines.com/en/ranking>). The version that we will be using is PostgreSQL 10.18.

2.1 Local Installation

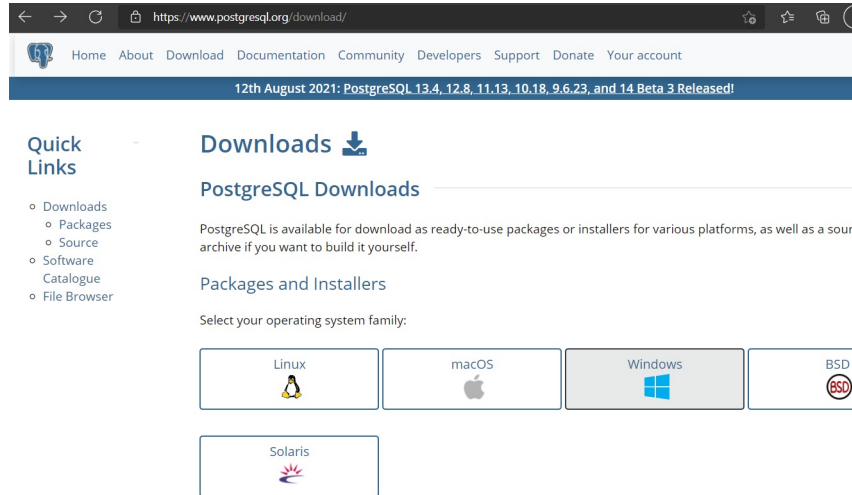
We request you all to install PostgreSQL 10.18 in your local machine. Please find the downloads at <https://www.postgresql.org/download/>. For Windows users, please go through the installation instructions given below.

Installing PostgreSQL in Windows 10

1. Go to: <https://www.postgresql.org/download/>



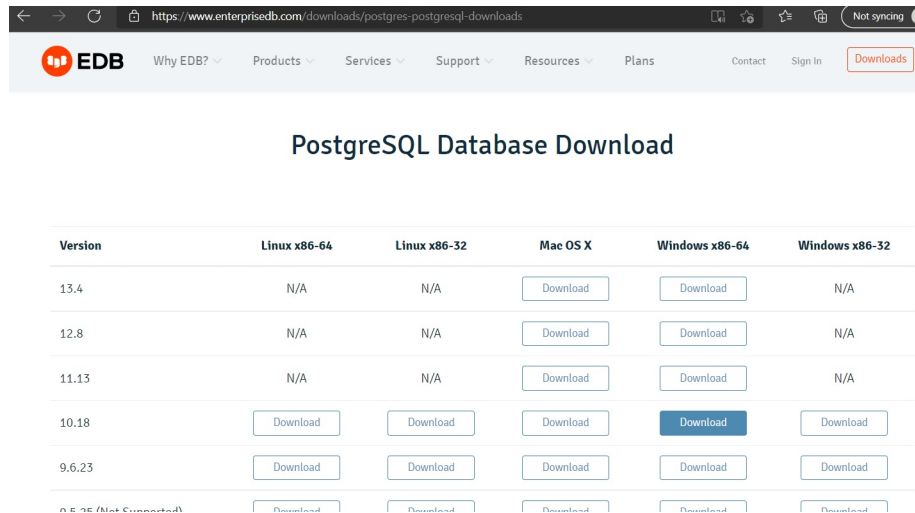
2. Click on Windows.



3. Then, you will be taken to the page as shown below:

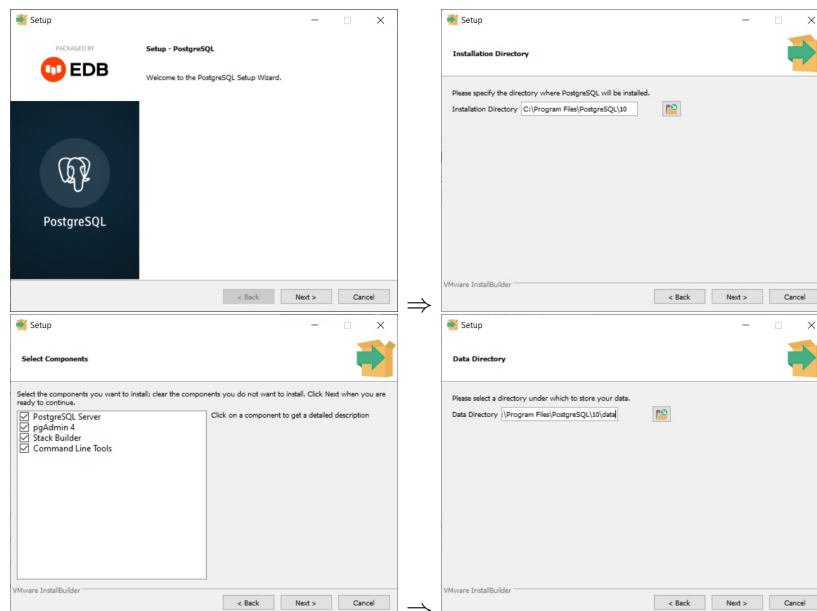


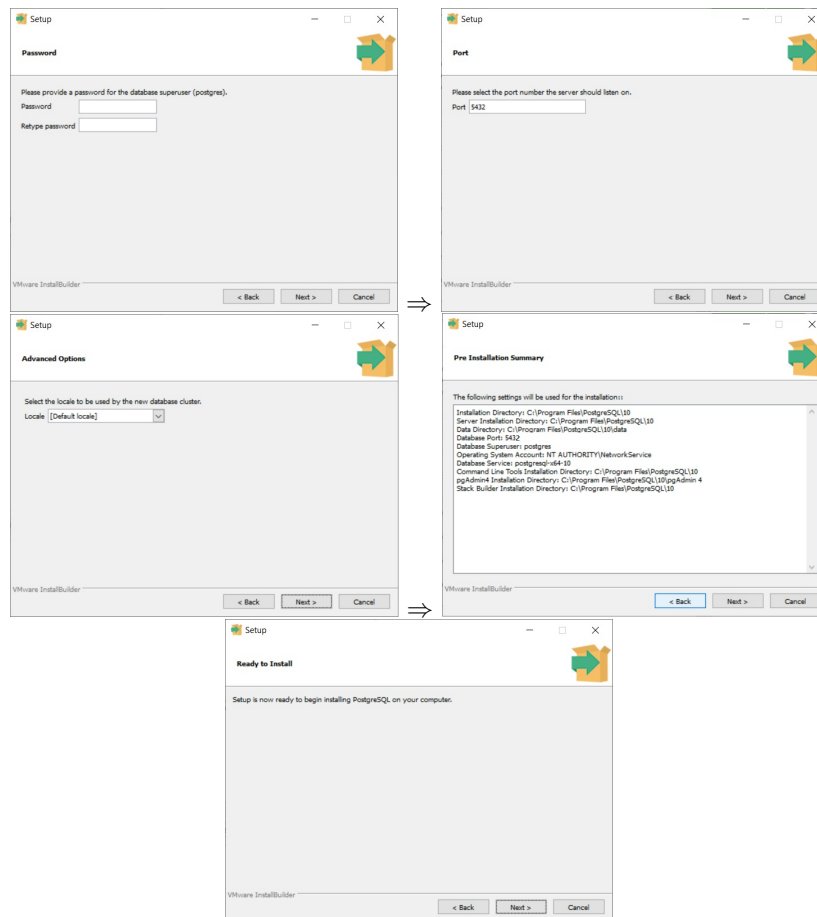
- Click on *Download the installer*, and you will see the page shown below:



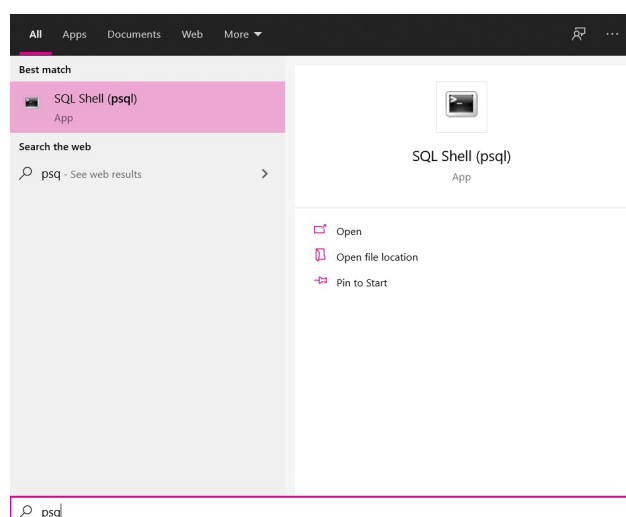
Version	Linux x86-64	Linux x86-32	Mac OS X	Windows x86-64	Windows x86-32
13.4	N/A	N/A	Download	Download	N/A
12.8	N/A	N/A	Download	Download	N/A
11.13	N/A	N/A	Download	Download	N/A
10.18	Download	Download	Download	Download	Download
9.6.23	Download	Download	Download	Download	Download
9.6.25 (Not Supported)	Download	Download	Download	Download	Download

- Click the download button for PostgreSQL 10.18, Windows x86-64 (or Windows x86-32, based on the architecture).
- Save the postgresql-10.18-1-windows-x64.exe in a folder.
- Double-click the .exe file to proceed with the installation.
- Follow the instructions on the screen. The below screenshots provide a guideline.



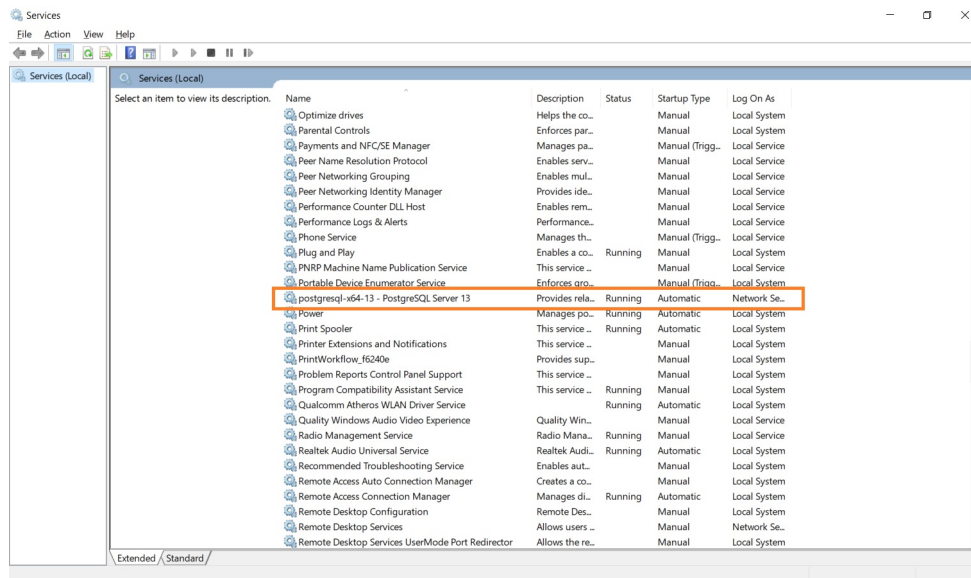


Accessing the local database: To access the PSQL prompt, search for psql. Click on the **SQL Shell (psql)**. See the screenshot below.



Please watch Tutorial 2.3 to connect to the database server.

Note: The service **postgresql-x64-13** must be running inside Control Panel\All Control Panel Items\Administrative Tools\Services for you to be able to access the database server. See the screenshot below.



Installing PostgreSQL in Ubuntu and macOS

For installation of PostgreSQL in Ubuntu, please follow the instructions in <https://www.postgresql.org/download/linux/ubuntu/>

OR

<https://phoenixnap.com/kb/how-to-install-postgresql-on-ubuntu>

For installing PostgreSQL in macOS, please follow the instructions in:

<https://www.postgresql.org/download/macosx/>

OR

<https://www.postgresqltutorial.com/install-postgresql-macos/>

Once the database server is installed, you can access the server using the SQL Shell (PSQL) or the GUI Interface provided by pgAdmin (<https://www.pgadmin.org/>).

2.2 Online database hosting

Although we strongly recommend that you install the PostgreSQL DB locally, there are many PostgreSQL database hosting platforms on which you can host your database for free. Some examples are:

- Heroku (<https://www.heroku.com/postgres>)
- ElephantSQL (<https://www.elephantsql.com/>)
- HelioHost (<https://www.heliohost.org/>)

- AmazonRDS (<https://aws.amazon.com/rds/postgresql/>)

Detailed help on configuring your database is available on the respective web-pages. We may be unable to give you guidance on each such service, understandably, from a practical perspective. Once you host the database in any of these platforms, you may access it via your local installation, or you could use front end tools like PopSQL (<https://popsql.com/>) or DBeaver (<https://dbeaver.io/>).

3 Live Sessions

3.1 Activity Sessions

In these sessions, we will address the Activity Questions for the given week. During these sessions, we plan to provide extra support to beginner level learners of DBMS. However, all students are welcome to join.

3.2 Practice/Solve With Instructor Sessions

In these sessions, we will address questions from Practice Assignments. There will be questions for which detailed solutions have not been provided in the solution document. We will pose such questions as ‘Solve With the Instructor’ questions and lead you through solving those. We request your cooperation in having familiarized yourself with the questions, before coming for the session so that we can make the best utilization of the 2 hours.

3.3 Open Sessions

You are free to ask any questions from the week, except from the graded assignments whose submission deadlines are not yet over.

3.4 Revision Sessions

These will be planned as and when required.

4 Query-based Questions

4.1 Mode of submission of solution

As mentioned during the orientation sessions, for specific weeks from Week 2 onwards, we are planning to have query-based questions. We will provide databases as .tar files for you to load into your working database server. Detailed instructions on how to load the .tar files are provided in a PDF in the Additional Resources folder. There will be questions for which you have to write a query as the answer. For practice questions, you can get feedback on clicking *Test Run* button. However, it may not be possible to obtain exact feedback from the portal; hence it is advisable to run the query on your local installation, obtain the answers and then copy the query into a file (any filename is ok) and upload it as described in the portal. Note that the query should NOT have a semicolon at the end, to enable us to perform auto-grading of queries. The

graded questions will be auto-evaluated once the submission deadline is over.

Important: It is mandatory to submit the practice query-based questions in order for your graded query-based submissions to be considered for evaluation.

This is to ensure you are familiar with the working of the portal for the query-based questions.

5 Datasets

We are planning to use two important database schemas for the query-based questions. The .tar files and .sql files for these databases are available under **Datasets** on the portal. The details about the table structures (the database schema) are also given in detail.